



Lab Experiment 1

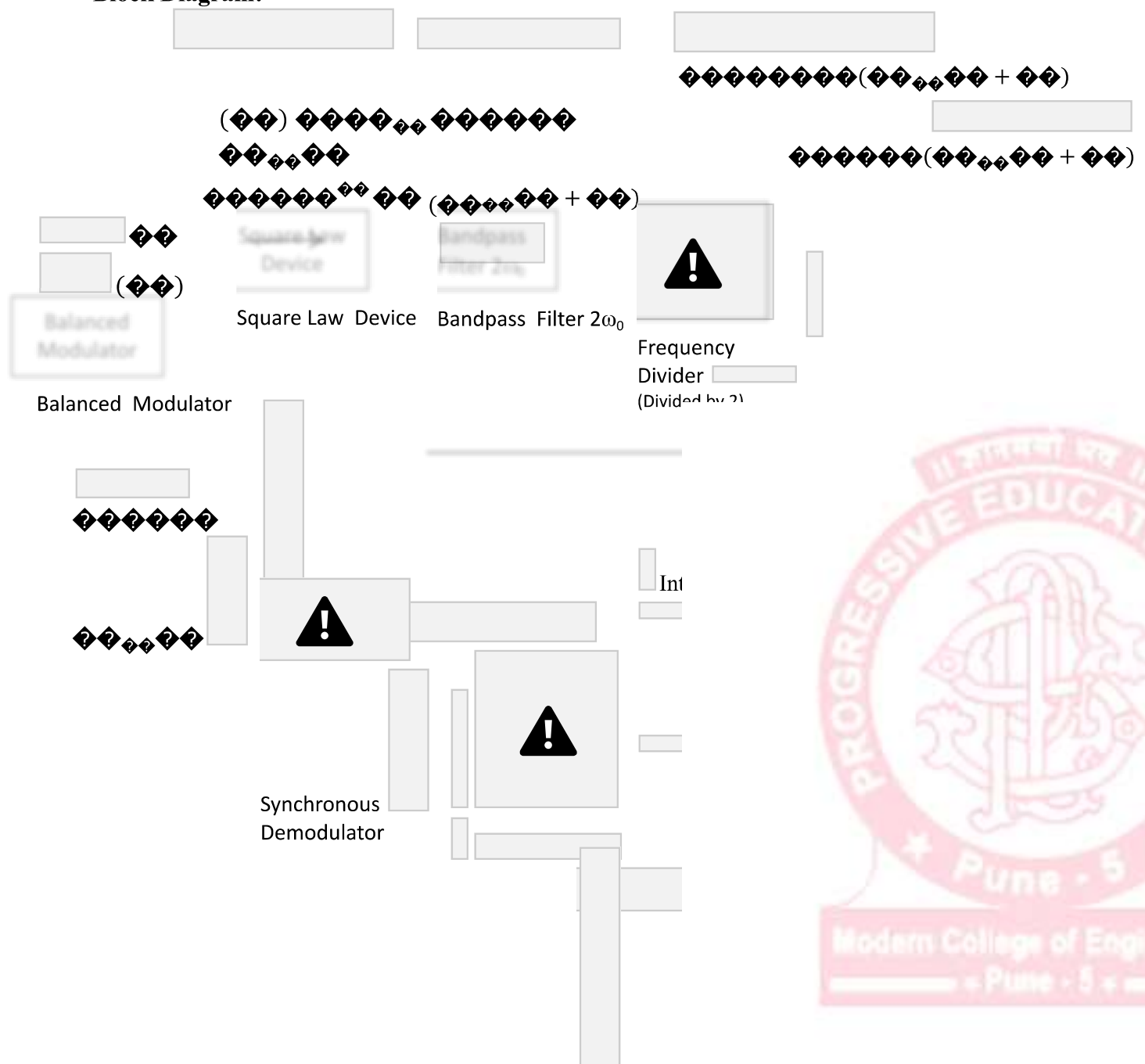
Experiment Title:

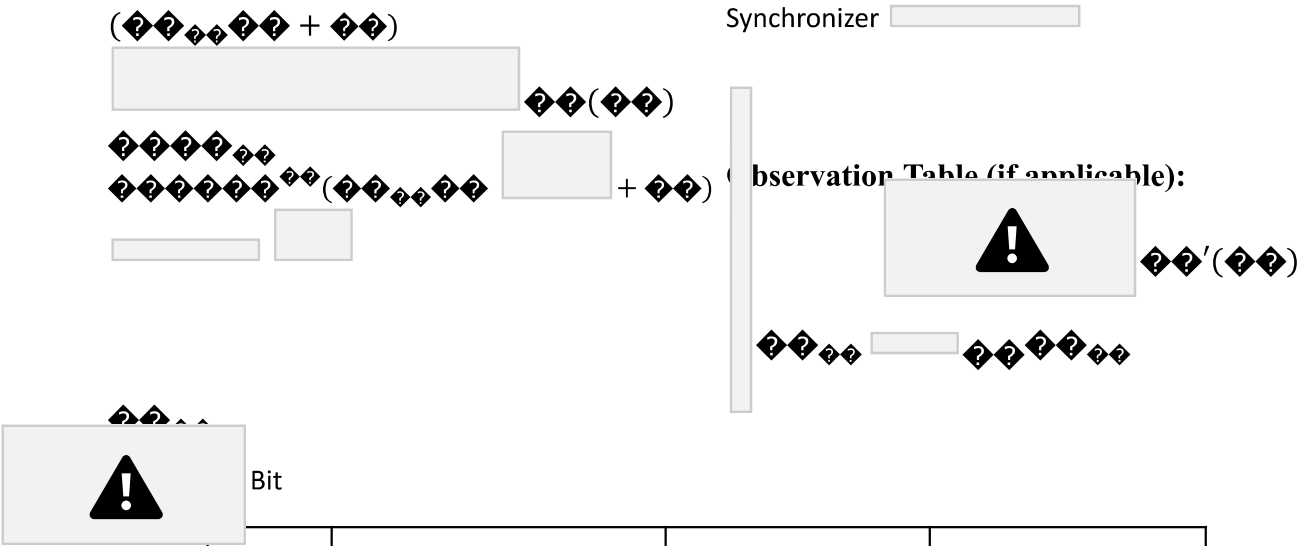
Generate and detect BPSK signals using suitable hardware and comment on the bandwidth.

Problem Statement:

In a digital communication system, binary data is transmitted by varying the phase of a carrier signal between two distinct states. Generate and detect signals using appropriate modulation technique and comment on the bandwidth required.

Block Diagram:





Sr. No.	Waveform Observed	Frequency (Hz)	Amplitude (V)
1	Input Data		
2	Carrier		
3	BPSK Signal		
4	Recovered Data		

Calculations:

1. Bandwidth = $2 * f_b =$ _____

- Draw waveforms for:
- 1. Input data
 - 2. Carrier signal
 - 3. BPSK signal
 - 4. Reconstructed carrier signal
 - 5. Reconstructed data

Spectrum:



Digital Communication Laboratory Program: T.Y. B.Tech. (E & TC)

Faculty of Science & Technology  PES's Modern College of Engineering Third Year
B.Tech. in Electronics and Telecommunication Engineering (2024 Pattern)

Conclusion:

Date of Performance		Date of Submission
Grade		
Sign of the Course Teacher with date		

